

2520030508 - WEEK 3

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Microsoft Windows [Version 10.0.26200.9188]
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C:\Windows\System32\cmd C:\Users\jithin\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin

C:\Users\jithin\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username jithinsurya_db_user
Enter password: ****
Current Mongosh Log ID: 6a89322969041f57525a779e
Connecting to:
  mongodb+srv://cluster0.eay9vj1.mongodb.net/?appName=mongosh+2.10.0
MongoServerSelectionError: 0x318009:error:0A000438:SSL routines:ssl3_read_bytes:tlsv1 alert internal error:openssl\ssl\record\rec_layer_s3.c:918:SSL alert number 80
  It looks like this is a MongoDB Atlas cluster. Please ensure that your Network Access List allows connections from your IP.

C:\Users\jithin\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username jithinsurya_db_user
Enter password: ****
Current Mongosh Log ID: 6a89322969041f57525a779e
Connecting to:
  mongodb+srv://cluster0.eay9vj1.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:
  8.0.29 (API Version 1)
Using Mongosh:
  2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

Atlas atlas-4upy6x-shard-0 [primary] test> show databases;
bookFlowDB      8.00 KiB
sample_mflix    143.55 MiB
schoolDB        164.00 KiB
test            132.00 KiB
admin           0 B
local           0 B
Atlas atlas-4upy6x-shard-0 [primary] test> use bookFlowDB;
switched to db bookFlowDB
Atlas atlas-4upy6x-shard-0 [primary] bookFlowDB> db.createCollection("book_metadata")
{ ok: 1 }
Atlas atlas-4upy6x-shard-0 [primary] bookFlowDB> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: { pages: 340, fileSize_MB: 4.2, drm: true, narrator: "K. Rao", runtime_min: 610 },
    avgRating: 4.6,
    reviews: [
      { member_id: "M1001", rating: 5, comment: "Excellent deep dive into digital systems design.", date: ISODate("2023-01-15") },
      { member_id: "M1042", rating: 4, comment: "Good but the Digital edition had formatting issues.", date: ISODate("2023-02-28") }
    ]
  },
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-18") },
      { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: { pages: 410, binding: "Sewn" },
    avgRating: 3.8,
    reviews: [
      { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
      { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
    ]
  },
  {
    title: "Quantum Horizons",
    author: "R. Menon",
    genre: "Science",
    publishedYear: 2023,
    formats: ["Digital", "Audiobook"],
    specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
      { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
    ]
  }
])
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a893229a60041f57525a779f'),
  '1': ObjectId('6a893229a60041f57525a779a')
}
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mongo@mongo> use atlas-atlas-4upyx-shard-0 [primary] BookFlowDB> db.book_metadata.find({ avgRating: { $gt: 4 } }, { title: 1, avgRating: 1, _id: 0 })
db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } }, { title: 1, formats: 1, _id: 0 })
db.book_metadata.find({ avgRating: { $gt: 4 }, formats: { $in: ["Digital"] } })
db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } }, { title: 1, formats: 1 })
db.book_metadata.find({ reviews.comment: { $regex: "Digital", $options: "i" } }, { title: 1, reviews.$: 1 })

{
  _id: ObjectId('6a89320a69041f57525a779f'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a89320a69041f57525a77a0'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-19T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a89320a69041f57525a77a2'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
]
atlas atlas-4upyx-shard-0 [primary] BookFlowDB> db.book_metadata.aggregate([
  { publishedYear: 2021,

```

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{
  _id: ObjectId('6a89320a69041f57525a77a2'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
]
atlas atlas-4upyx-shard-0 [primary] BookFlowDB> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: '$reviews' },
  { $group: {
    _id: '$title',
    avgReviewRating: { $avg: '$reviews.rating' },
    totalReviews: { $sum: 1 },
    author: { $first: '$author' }
  }},
  { $sort: { avgReviewRating: -1 } },
  { $project: {
    _id: 0,
    title: '$_id',
    author: 1,
    avgReviewRating: { $round: ['$avgReviewRating', 2] },
    totalReviews: 1
  }},
  {
    totalReviews: 2,
    author: 'R. Renon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandez',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'R. Renon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  }
]
publishedYear: 2021,

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    multiKeyPaths: { avgRating: [] } },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 1,
    direction: 'forward',
    indexBounds: { avgRating: [ '[min:0, 4]' ] },
    keysExamined: 3,
    seeks: 1,
    docsTested: 0,
    docsDropped: 0
  }
},
queryShapelHash: '17380C6D409D1958BA14FE356779B269D01F431795C41057B22D5D8FECABCA3A8',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$obj': 'Book/1a0C8'
},
serverInfo: {
  host: 'ac-pubsub-shard-00-02.eay9wjl.mongodb.net',
  port: 27017,
  version: '4.0.20',
  gitVersion: '593067c651f7393e2757234a0b13cb7e8622140'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 1m4857600,
  internalLookupFlagIndexReadaheadDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceDocMemoryBytes: 164857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitLockingMergeOnMongos: 0,
  internalQueryMaxAddToGetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'try_disable_retry',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'clusterTime': {
  clusterTime: Timestamp({ t: 1787376305, i: 36 }),
  signature: {
    hash: Binary.createFromBase64('PTPbXonIQncR3UHKiHrmlVnc=', 0),
    keyId: Long('761840689818992483')
  }
},
operationTime: Timestamp({ t: 1787376305, i: 36 })
},
Atlas atlas-4up6x-shard-0 [primary] Book/1a0C8
publishedYear: 2021

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